Valid Starting City

AlgoExpert · Medium · Greedy Algorithms

The problem

- Circular route; city i has fuel[i] gallons and is distances[i] miles from the next.
- Return the unique start index allowing a full loop.
- The start is the point after the lowest fuel balance.

Intuition

- The valid start is right after the point where the running fuel balance is lowest.

Approach

- 1. Track miles remaining around the loop
- 2. Find the city with the minimum balance
- 3. That city is the start

Java solution

```java
public class ValidStartingCity {
    public static int validStartingCity(int[] distances, int[] fuel, int mpg) {
        int miles = 0, minMiles = 0, start = 0;
        for (int i = 1; i < distances.length; i++) {
            miles += fuel[i - 1] * mpg - distances[i - 1];
            if (miles < minMiles) { minMiles = miles; start = i; }
        }
        return start;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- The valid start is right after the point where the running fuel balance is lowest.
- Complexity: O(n)